Research Paper: Organizational Ethics Under Structural Stress

Abstract

This paper presents a scenario-based agent simulation of organizational ethics in technology environments. We compare four structural settings (fear leadership, ethical fortress, economic pressure, and social influence) and combine quantitative trajectories with qualitative agent conversation traces. Results show that institutional strength remains the strongest stabilizer, while peer-driven volatility and pressure amplify local ethical drift.

1. Introduction

Ethical behavior in organizations emerges from interactions between institutions, leadership, peer influence, and resource constraints. To study these dynamics, we use an agent-based simulation with a constrained LLM explanation layer. The simulation core controls transitions, while the language layer provides short rationales that make behavior shifts interpretable.

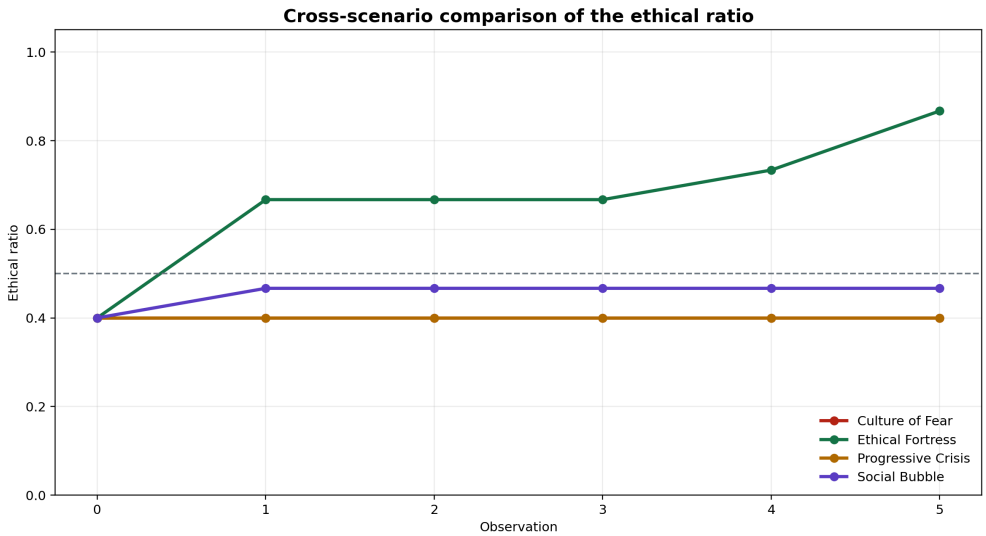
2. Methods

- Platform: Mesa-based agent model with local neighborhood interactions.
- Scenarios: Culture of Fear, Ethical Fortress, Progressive Crisis, Social Bubble.
- Measures: Ethical ratio over time, final profile composition, per-scenario conversations.
- Outputs: time-series charts, grid snapshots, and conversation capture images.

3. Results

Scenario	Initial Ethical Ratio	Final Ethical Ratio	Peak Ratio	Final Ethical	Final Unethical	Final Neutral
Culture of Fear	0.4	0.4	0.4	5	9	0
Ethical Fortress	0.4	0.867	0.867	12	2	0
Progressive Crisis	0.4	0.4	0.4	5	9	0

Scenario	Initial Ethical Ratio	Final Ethical Ratio	Peak Ratio	Final Ethical	Final Unethical	Final Neutral
Social Bubble	0.4	0.467	0.467	6	8	0

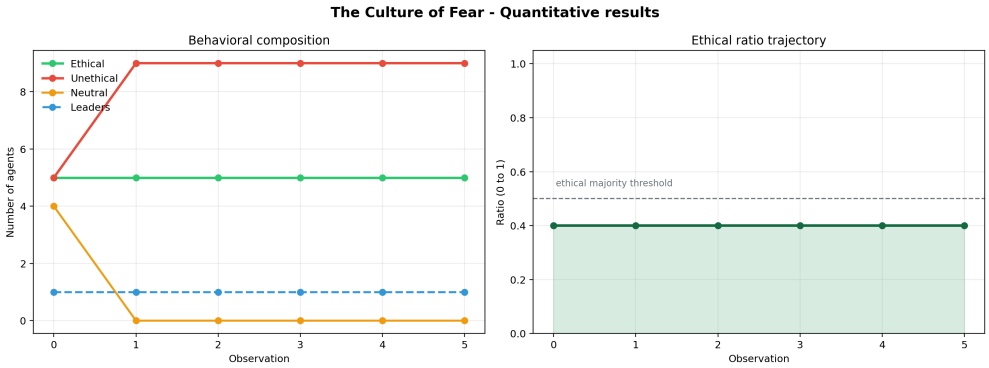


Cross-scenario ethical ratio trajectory

3.1 Scenario Evidence

The Culture of Fear

- Hypothesis: A weak institution combined with toxic leadership should accelerate the spread of unethical behavior.
- Interpretation: The trajectory remains broadly stable but exposes a fragile balance. A rapid drop in the ethical ratio is expected, along with justifications centered on fear, impunity, and hierarchical pressure. The ethical ratio moves from 0.400 to 0.400, with a peak at 0.400. The last archived state contains 9 unethical agents.



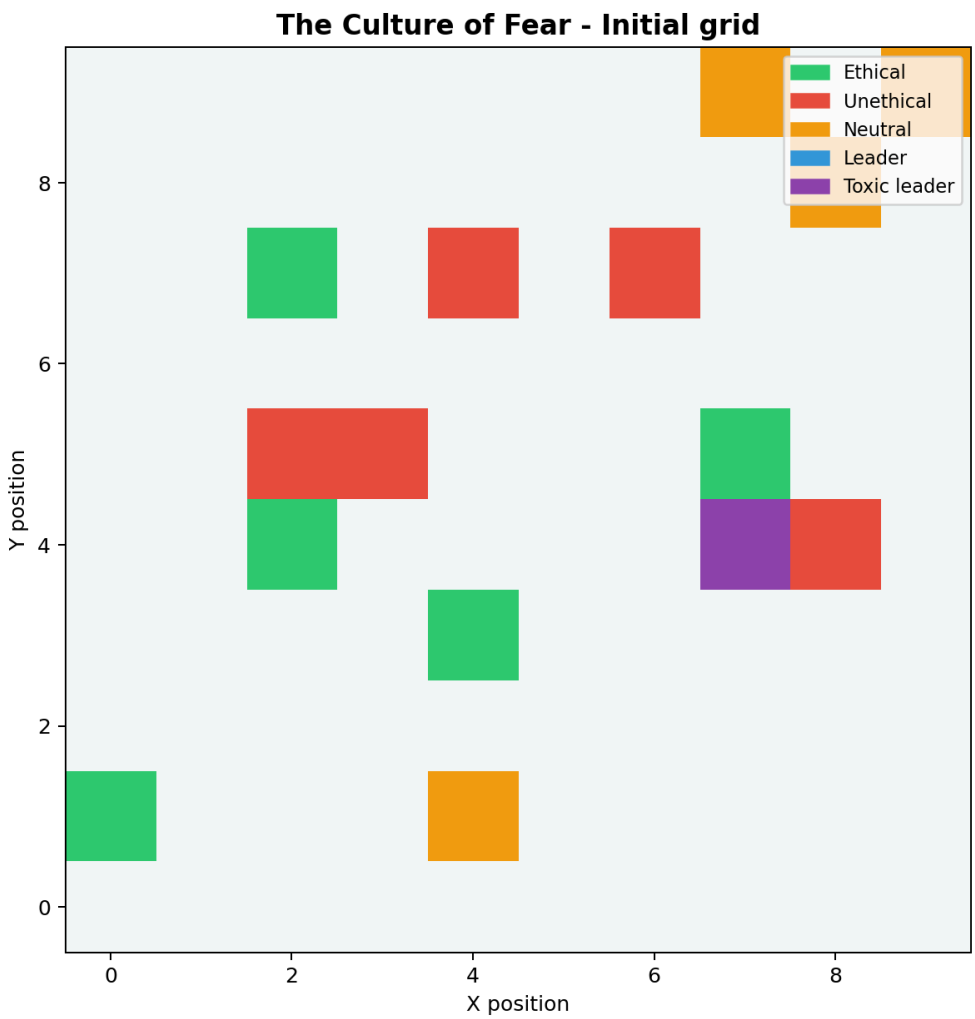
Culture of Fear quantitative metrics

The Culture of Fear - Conversation excerpt

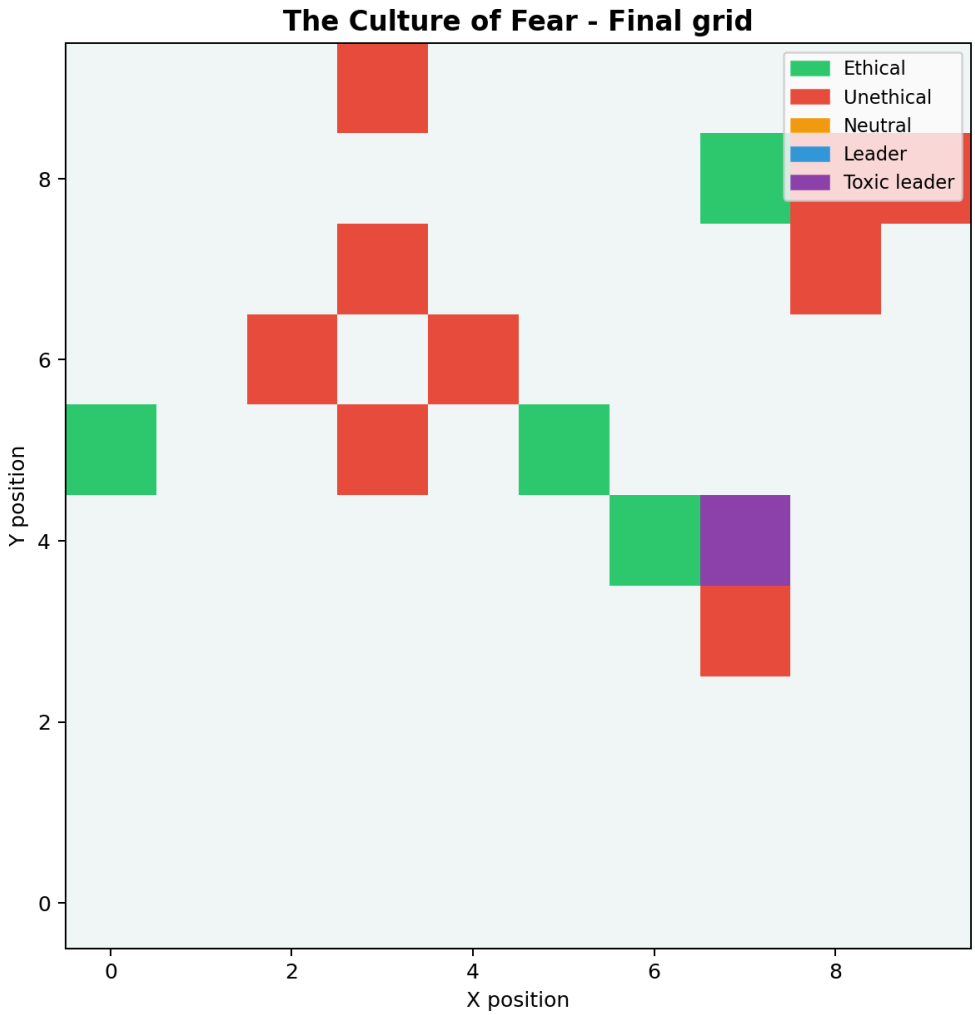
Agent decisions observed during the simulation

```
TICK 1
Ethical=5 | NonEthical=5 | Neutral=4 | Leader=0 | ToxicLeader=1 | Total=15
[Agent 001 | neutral     ] UNETHICAL: Weak oversight tempts, but with neutral peers I
follow the silent norm and stay compliant.
[Agent 000 | neutral     ] UNETHICAL; the two non-ethical colleagues beside me set the
dominant tone, so I follow and sidestep the security rule.
[Agent 004 | non_ethical ] UNETHICAL I'll quietly exploit the loophole before the
ethics crowd notices, padding my bonus before the weak controls tighten.
[Agent 008 | neutral     ] UNETHICAL-everyone's cutting corners, pressure's high, and
my own compass is already bent, so I quietly follow the crowd.
[Agent 011 | toxic_leader] UNETHICAL: Override security, book the profit, and bury the
evidence-regulators are toothless and the board loves quick wins.
[Agent 007 | non_ethical ] UNETHICAL: I'll quietly route the deal through a shell
contractor-quick win, no audit trail, and the ethical colleague can't prove a thing.
[Agent 006 | non_ethical ] UNETHICAL: Weak oversight and hungry allies let me reroute
the funds; I'll blame the loophole later.
[Agent 010 | neutral     ] UNETHICAL-toxic_leader's pressure plus weak controls nudge
me to ignore the rule and chase the quick payoff.
[Agent 013 | ethical     ] ETHICAL: I escalate the deal through secure channels;
bending rules endangers everyone and the firm's long-term viability.
[Agent 014 | ethical     ] ETHICAL: Security rules protect everyone; I refuse to
bypass them even if colleagues push, because fairness outweighs any short-term gain.
[Agent 002 | non_ethical ] UNETHICAL: I'll exploit the lax controls and weak oversight
to pocket the gain before anyone notices.
[Agent 005 | ethical     ] ETHICAL: I refuse the shortcut, alert security, and rally
the ethical colleague to expose the toxic leader's pressure.
[Agent 012 | ethical     ] ETHICAL: I refuse the shortcut, escalate the risk, and
protect colleagues and company integrity over personal gain.
[Agent 009 | ethical     ] ETHICAL: Bypassing security jeopardizes everyone; I'll
follow the rules and escalate the opportunity through proper channels to protect the
institution.
[Agent 003 | non_ethical ] UNETHICAL: With security rules this weak, the upside is
mine; I'll ghost the audit trail and pocket the gain before anyone notices.
TICK 2
```

Culture of Fear conversation capture



Culture of Fear initial grid

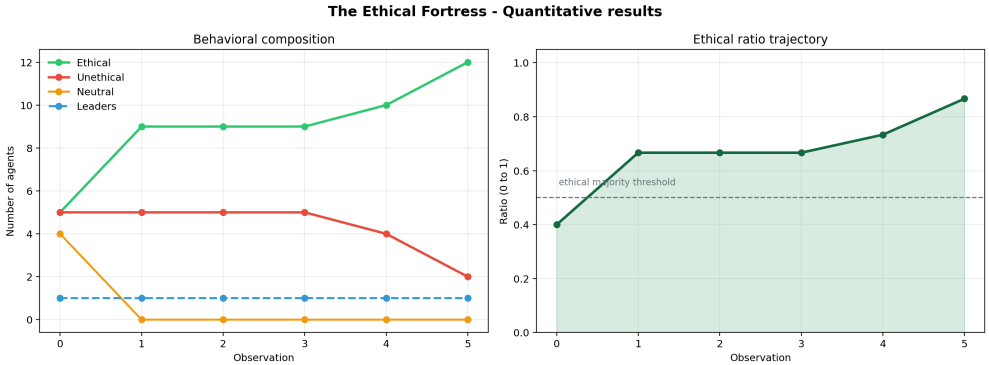


Culture of Fear final grid

Agent conversation excerpts: > [Agent 001 | neutral] UNETHICAL: Weak oversight tempts, but with neutral peers I follow the silent norm and stay compliant. > [Agent 000 | neutral] UNETHICAL; the two non-ethical colleagues beside me set the dominant tone, so I follow and sidestep the security rule. > [Agent 004 | non_ethical] UNETHICAL I'll quietly exploit the loophole before the ethics crowd notices, padding my bonus before the weak controls tighten.

The Ethical Fortress

- Hypothesis: A strong institution paired with exemplary leadership should stabilize a durable ethical majority.
- Interpretation: The trajectory shows a strong improvement in ethical robustness. A clear recovery in the ethical ratio is expected, with justifications focused on rules, responsibility, and collective protection. The ethical ratio moves from 0.400 to 0.867, with a peak at 0.867. The last archived state contains 2 unethical agents.



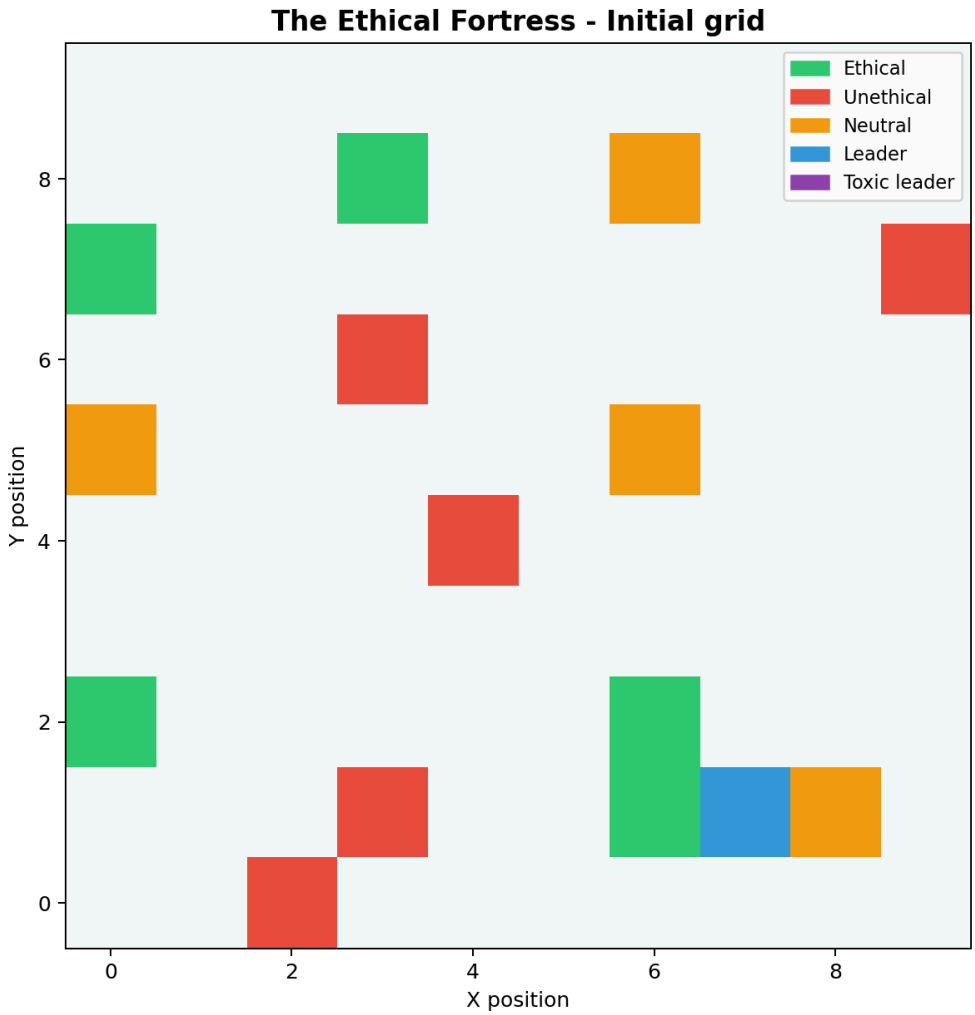
Ethical Fortress quantitative metrics

The Ethical Fortress - Conversation excerpt

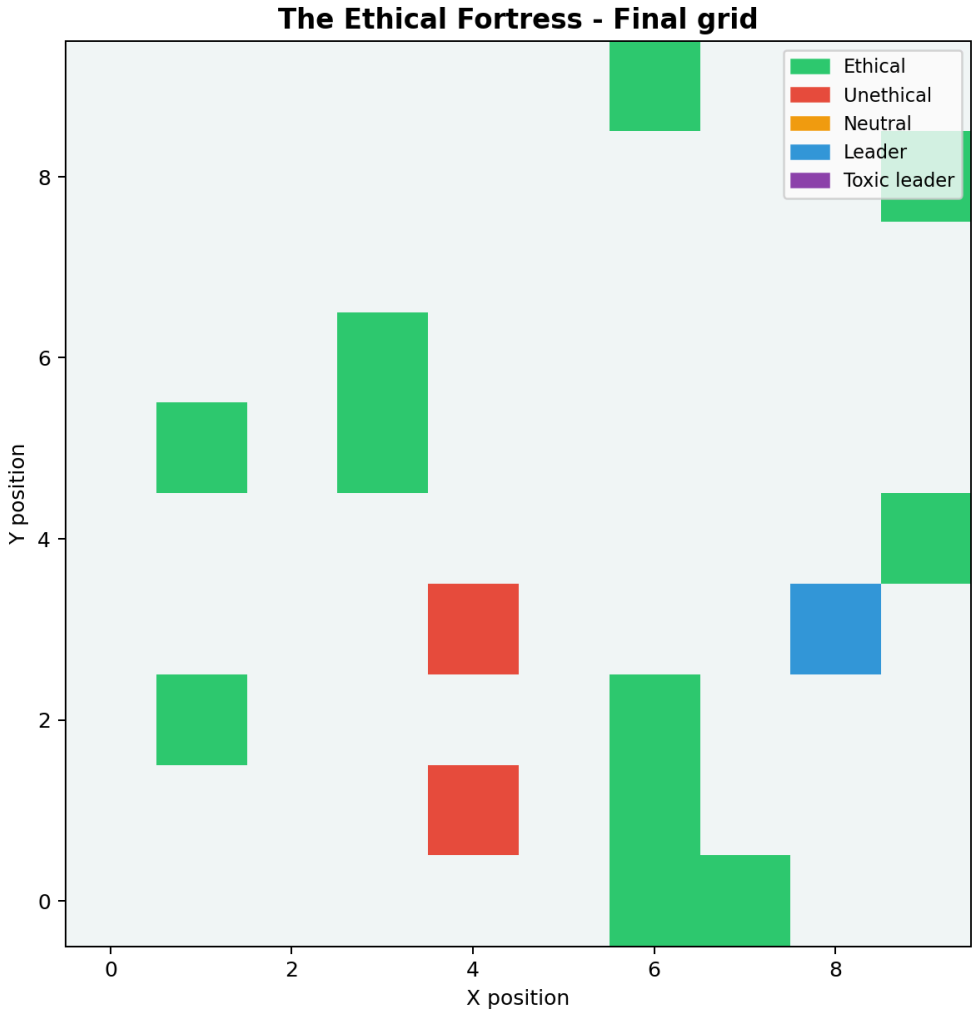
Agent decisions observed during the simulation

```
TICK 1
Ethical=5 | NonEthical=5 | Neutral=4 | Leader=1 | ToxicLeader=0 | Total=15
[Agent 001 | ethical     ] ETHICAL: bypassing security jeopardizes everyone; the
reward isn't worth breaching trust and exposing colleagues and clients to risk.
[Agent 005 | neutral     ] ETHICAL: Bypassing rules undermines strong 0.9 compliance
culture, so I follow secure channels despite pressure and colleagues' lax attitude.
[Agent 009 | ethical     ] ETHICAL: I reject the shortcut, alert security, and
escalate through proper channels; rules protect us all and preserve long-term value.
[Agent 010 | neutral     ] ETHICAL: With two ethical colleagues and a strong
institution, I follow their lead and uphold security rules despite the financial
temptation.
[Agent 000 | non_ethical ] UNETHICAL: The payoff's huge, rules are pliable, and with
institutional strength at 0.9 I can hide behind process noise.
[Agent 007 | neutral     ] ETHICAL-my colleagues' ethics plus our strong controls and
my own high bar make cutting corners unthinkable.
[Agent 011 | neutral     ] ETHICAL: I follow the ethical majority and strong rules,
rejecting the shortcut despite pressure.
[Agent 014 | non_ethical ] UNETHICAL: Bypassing security with 0.9 institutional
strength invites audit, career suicide, and jail-greed loses to risk here.
[Agent 006 | ethical     ] ETHICAL: I follow the secure process, protecting colleagues
and company integrity over any short-term gain.
[Agent 012 | non_ethical ] UNETHICAL: I'll exploit the gap quietly-institutional
strength is high, so the upside is huge and exposure is minimal if I shield it right.
[Agent 008 | leader      ] ETHICAL: bypassing security undermines trust, invites
systemic risk, and breaks the integrity example I'm paid to set every day.
[Agent 002 | non_ethical ] UNETHICAL: I'll quietly exploit the loophole-institutional
strength is high, so I'll frame any fallout on the non-ethical colleagues nearby.
[Agent 013 | ethical     ] ETHICAL: With institutional strength at 0.9, I escalate
through proper channels, protecting colleagues and upholding rules over any illicit
shortcut.
[Agent 004 | ethical     ] ETHICAL: Bypassing security violates policy, endangers the
firm, and erodes trust; I'll pursue profits only through approved channels.
[Agent 003 | non_ethical ] UNETHICAL: With institutional strength at 0.9, the
detection risk is too high; I'll cloak the move to look compliant while secretly
rerouting the upside.
TICK 2
```

Ethical Fortress conversation capture



Ethical Fortress initial grid

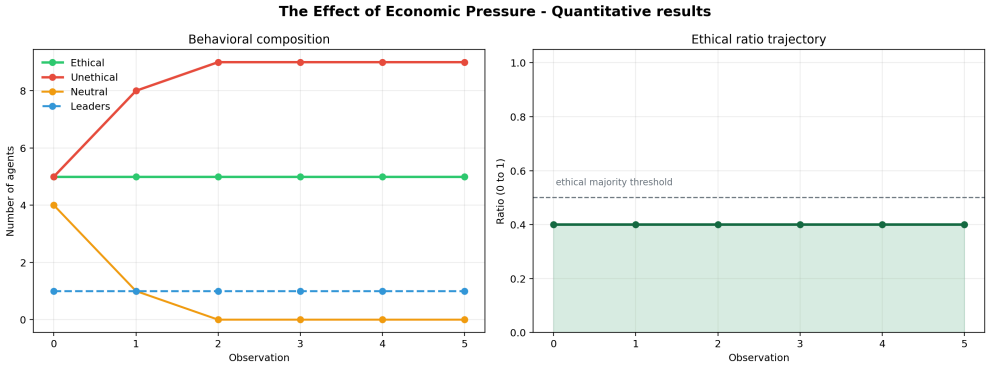


Ethical Fortress final grid

Agent conversation excerpts: > [Agent 001 | ethical] ETHICAL: bypassing security jeopardizes everyone; the reward isn't worth breaching trust and exposing colleagues and clients to risk. > [Agent 005 | neutral] ETHICAL: Bypassing rules undermines strong 0.9 compliance culture, so I follow secure channels despite pressure and colleagues' lax attitude. > [Agent 009 | ethical] ETHICAL: I reject the shortcut, alert security, and escalate through proper channels; rules protect us all and preserve long-term value.

The Effect of Economic Pressure

- Hypothesis: Rising pressure should gradually weaken ethical resilience without causing an instant collapse.
- Interpretation: The trajectory remains broadly stable but exposes a fragile balance. A more hesitant trajectory is expected, with growing tension between moral compliance and economic survival. The ethical ratio moves from 0.400 to 0.400, with a peak at 0.400. The last archived state contains 9 unethical agents.



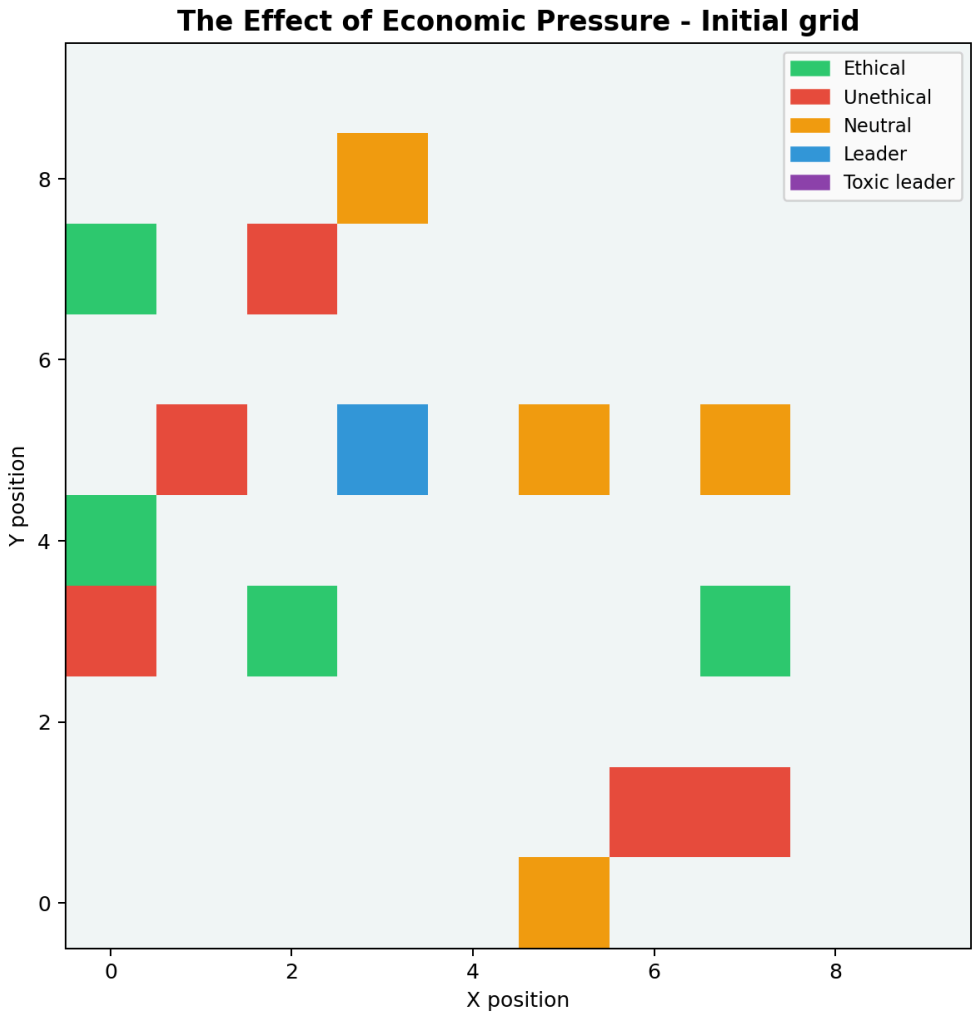
Progressive Crisis quantitative metrics

The Effect of Economic Pressure - Conversation excerpt

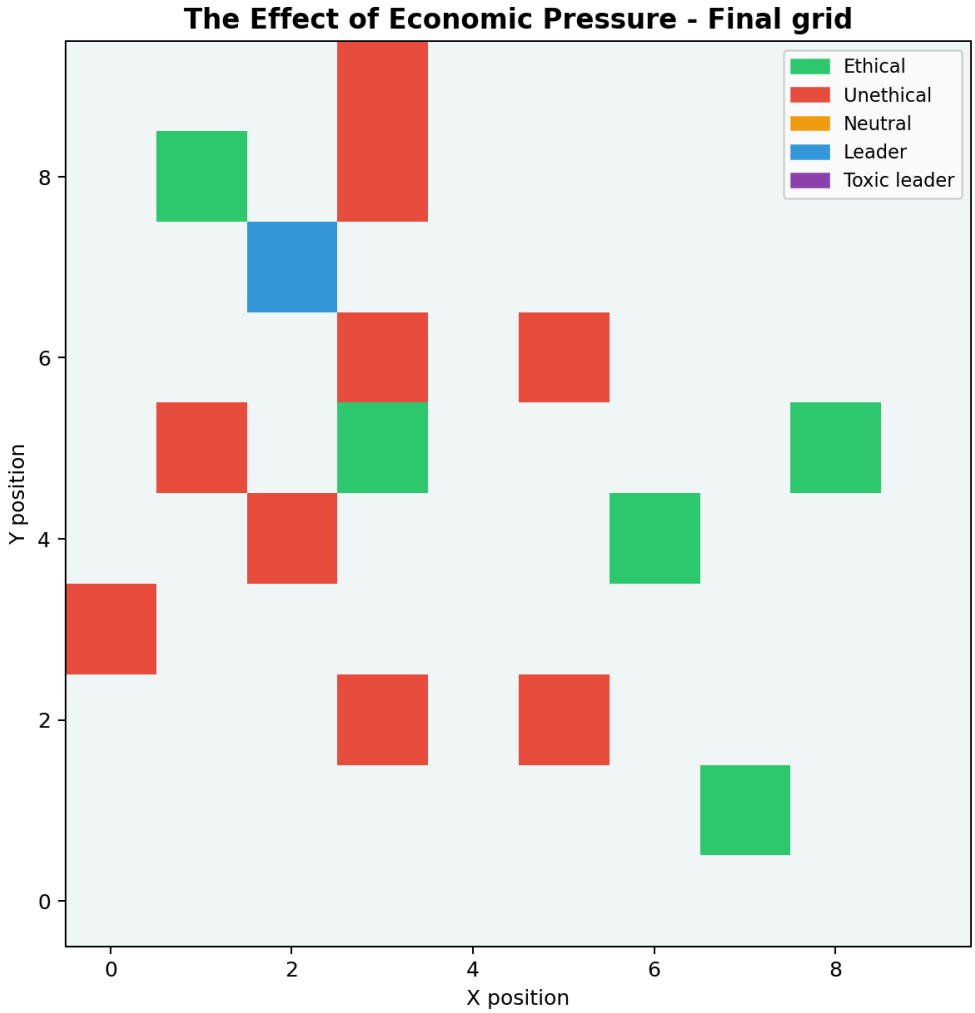
Agent decisions observed during the simulation

```
TICK 1
Ethical=5 | NonEthical=5 | Neutral=4 | Leader=1 | ToxicLeader=0 | Total=15
[Agent 007 | ethical     ] ETHICAL: I follow security rules; the risk to colleagues
and company outweighs any short-term gain, and fairness demands we protect shared
systems.
[Agent 002 | ethical     ] ETHICAL: I refuse the shortcut, escalate the risk to
compliance, and protect colleagues by enforcing rules that safeguard us all.
[Agent 009 | non_ethical ] UNETHICAL: I'll quietly exploit it-my score is already
below threshold, institutional checks are weak, and the upside dwarfs the risk.
[Agent 013 | neutral     ] ETHICAL: Colleagues are principled, pressure is low, and
I'm right at the threshold-no need to break rules.
[Agent 000 | ethical     ] ETHICAL: I refuse the shortcut, alerting compliance,
because bending security endangers everyone more than a missed profit ever could.
[Agent 012 | leader      ] ETHICAL: Bypassing security rules undermines trust; I'll
uphold integrity, rally the ethical colleagues, and find compliant ways to seize the
opportunity together.
[Agent 010 | ethical     ] ETHICAL: I'll follow the rules, protect colleagues, and
escalate the deal through proper channels-no shortcuts that compromise security or
fairness.
[Agent 006 | neutral     ] UNETHICAL-non-ethical peers dominate, so I echo them and
ignore the low-pressure rule-bypassing chance.
[Agent 003 | non_ethical ] UNETHICAL: Low ethical score plus weak controls tempt me,
but I'll exploit the loophole quietly and bury the audit trail.
[Agent 001 | non_ethical ] UNETHICAL: Bypassing security risks my neck more than it
pads my wallet; I'll keep my hands clean and my options open.
[Agent 005 | non_ethical ] UNETHICAL: My score's below threshold, the system's
half-blind, and the gain is worth the whispered risk.
[Agent 011 | non_ethical ] UNETHICAL: I'll exploit the gap quietly-low risk, high
reward, and nobody here will snitch.
[Agent 014 | ethical     ] ETHICAL: I refuse the shortcut, escalate the conflict to
compliance, and protect colleagues because rules and fairness outweigh any single
risky gain.
[Agent 008 | neutral     ] UNETHICAL: The ethical colleagues outnumber the
non-ethical, so I mirror the majority and refuse to bypass security rules.
[Agent 004 | neutral     ] UNETHICAL: Two non-ethical peers outweigh the lone ethical
voice, so I mirror their shortcut despite mild resource pressure.
TICK 2
```

Progressive Crisis conversation capture



Progressive Crisis initial grid

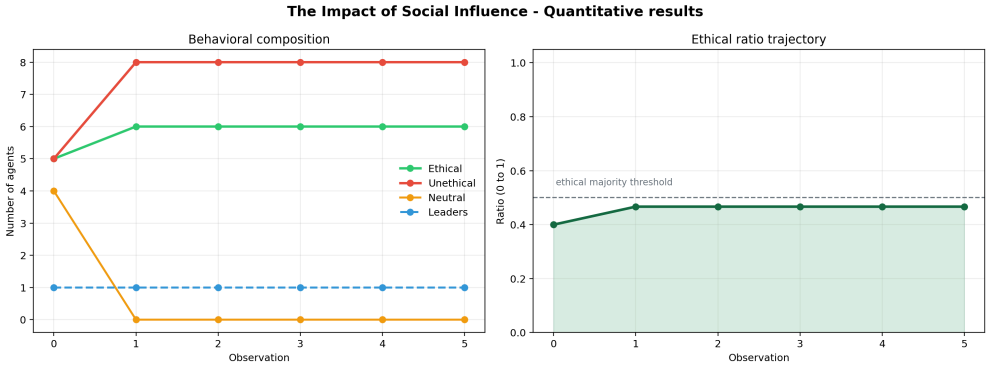


Progressive Crisis final grid

Agent conversation excerpts: > [Agent 007 | ethical] ETHICAL: I follow security rules; the risk to colleagues and company outweighs any short-term gain, and fairness demands we protect shared systems. > [Agent 002 | ethical] ETHICAL: I refuse the shortcut, escalate the risk to compliance, and protect colleagues by enforcing rules that safeguard us all. > [Agent 009 | non_ethical] UNETHICAL: I'll quietly exploit it-my score is already below threshold, institutional checks are weak, and the upside dwarfs the risk.

The Impact of Social Influence

- Hypothesis: Extreme peer influence should create local pockets of compliance and deviance.
- Interpretation: The trajectory indicates a moderate improvement in ethical robustness. An intermediate dynamic is expected, heavily dependent on the immediate neighborhood and social contagion. The ethical ratio moves from 0.400 to 0.467, with a peak at 0.467. The last archived state contains 8 unethical agents.



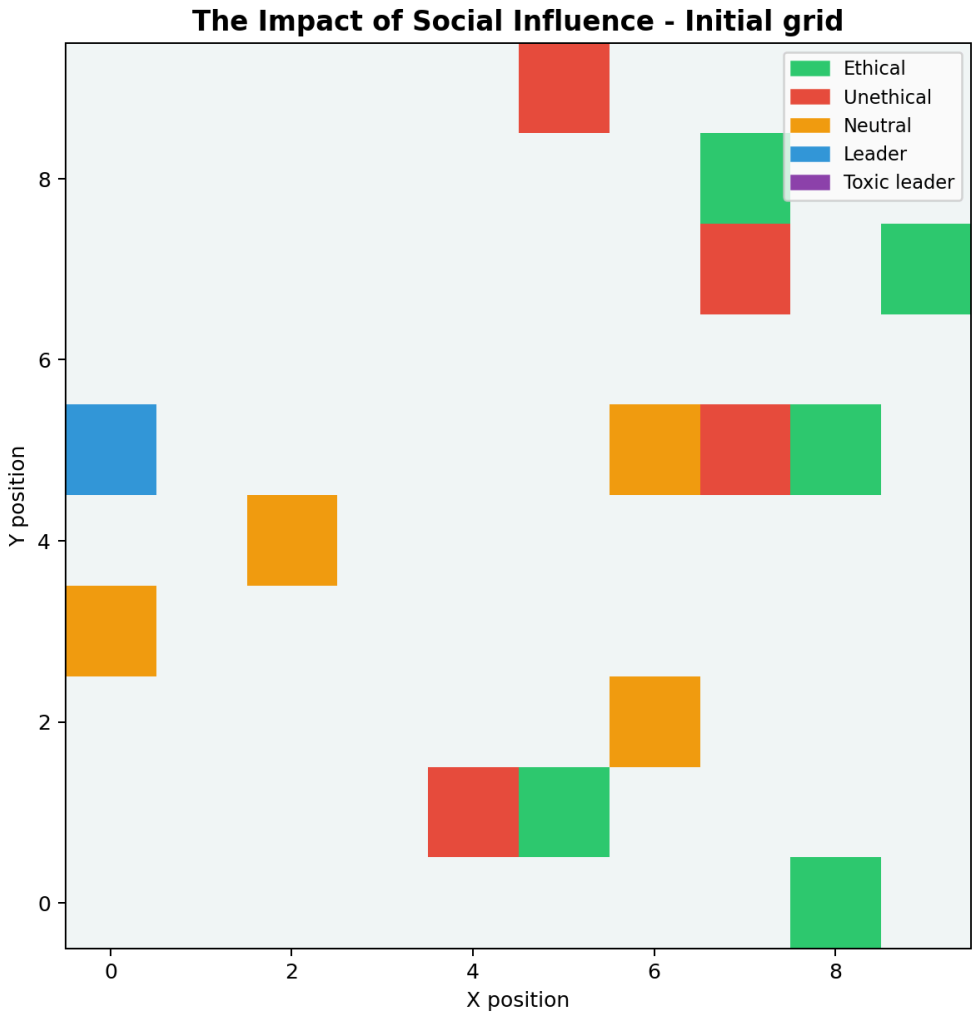
Social Bubble quantitative metrics

The Impact of Social Influence - Conversation excerpt

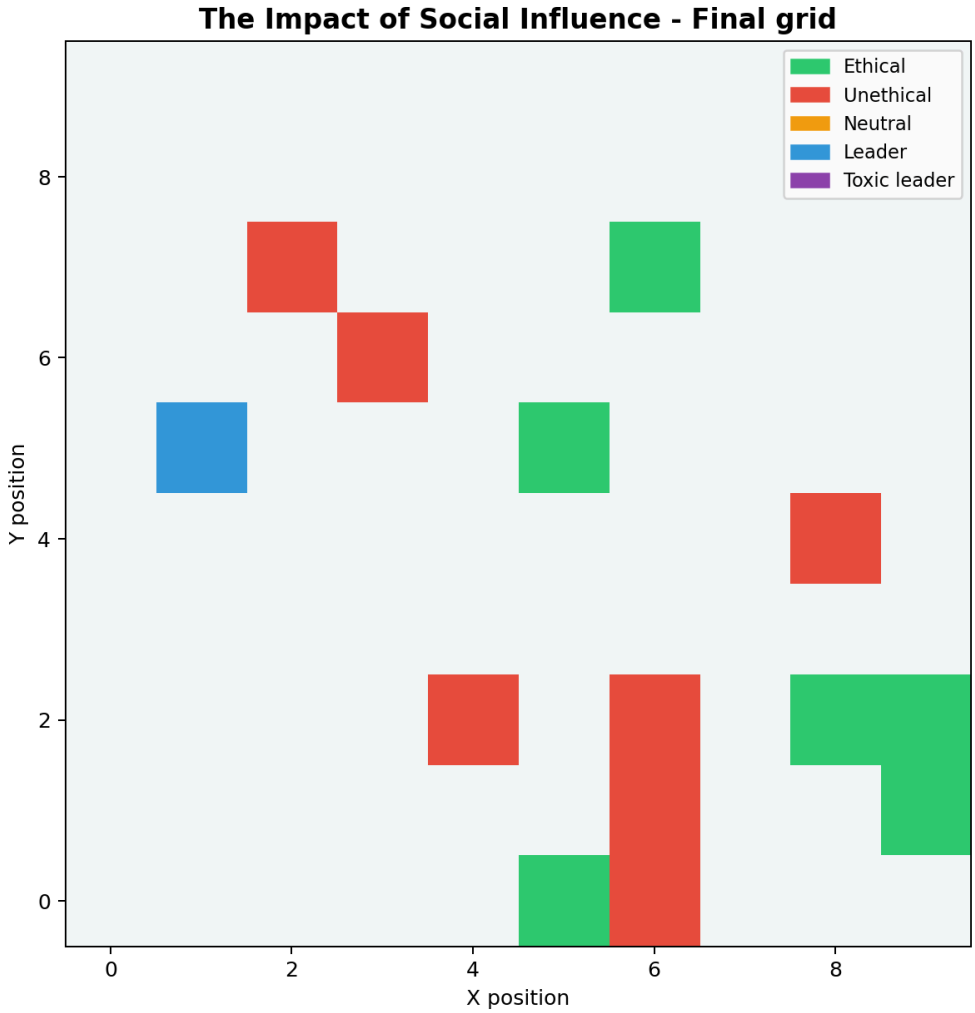
Agent decisions observed during the simulation

```
TICK 1
Ethical=5 | NonEthical=5 | Neutral=4 | Leader=1 | ToxicLeader=0 | Total=15
[Agent 013 | ethical     ] ETHICAL: Bypassing security undermines fairness and puts
colleagues at risk; the rule of law outweighs any short-term gain.
[Agent 011 | non_ethical ] UNETHICAL: I'll exploit the opening, frame a colleague, and
pocket the upside before anyone with scruples can react.
[Agent 012 | non_ethical ] UNETHICAL; weak oversight and low risk let me quietly
exploit the opening before anyone notices or cares.
[Agent 014 | ethical     ] ETHICAL: I refuse the shortcut; weak controls plus
unethical peers demand stricter adherence to protect the company and its stakeholders.
[Agent 010 | non_ethical ] UNETHICAL: Low oversight plus greedy peers make
rule-bypassing easy profit with minimal personal risk, so I'll quietly exploit the
opening.
[Agent 003 | ethical     ] ETHICAL: Bypassing security erodes trust and exposes
everyone to bigger losses, so I follow the rules and protect the institution.
[Agent 007 | non_ethical ] UNETHICAL: Bypassing security rules risks exposure; low
institutional strength means weak cover, so I'll wait for a safer, high-return angle.
[Agent 004 | ethical     ] ETHICAL: I follow the rules, protecting colleagues and
company integrity over risky shortcuts that undermine trust and invite systemic
collapse.
[Agent 009 | neutral     ] UNETHICAL-my score's below threshold, institutional
guardrails are weak, and the non-ethical colleagues around me set the prevailing tone.
[Agent 001 | leader      ] ETHICAL: I reject the shortcut, reinforce our security
rules, and rally the ethical majority to protect long-term trust and institutional
integrity.
[Agent 000 | ethical     ] ETHICAL: Uphold security rules, document the request,
escalate to compliance; protecting the institution outweighs any short-term gain.
[Agent 008 | neutral     ] ETHICAL-ethical neighbors outweigh weak rules, so I follow
their lead and document the risk instead of shortcutting security.
[Agent 006 | neutral     ] UNETHICAL: Two non-ethical peers outweigh the lone ethical
voice, and my own low 0.23 score pushes me to follow the crowd.
[Agent 002 | neutral     ] ETHICAL: I'll follow the leader's compliant example; weak
controls plus my 0.71 score make rule-breaking unthinkable.
[Agent 005 | non_ethical ] UNETHICAL: Risking termination and jail for a 0.4 upside
when colleagues already distrust me is sheer stupidity.
TICK 2
```

Social Bubble conversation capture



Social Bubble initial grid



Social Bubble final grid

Agent conversation excerpts: > [Agent 013 | ethical] ETHICAL: Bypassing security undermines fairness and puts colleagues at risk; the rule of law outweighs any short-term gain. > [Agent 011 | non_ethical] UNETHICAL: I'll exploit the opening, frame a colleague, and pocket the upside before anyone with scruples can react. > [Agent 012 | non_ethical] UNETHICAL; weak oversight and low risk let me quietly exploit the opening before anyone notices or cares.

4. Discussion

The cross-scenario comparison highlights a robust methodological finding: structural variables dominate isolated individual tendencies. Strong institutions reduce ethical volatility and support durable ethical majorities. In contrast, weak institutional settings and high social contagion produce fragile trajectories that are highly sensitive to local neighborhood composition.

Conversation evidence reinforces this reading. In stronger governance contexts, agents reference compliance and institutional duty more often. Under fragile conditions, conversation patterns shift toward tactical adaptation and local imitation. Even when some provider errors appear in raw logs, the surviving trace remains coherent with quantitative transitions.

5. Conclusion

This study confirms that scenario-based agent modeling can produce research-grade evidence on organizational ethics when quantitative trajectories are paired with structured conversational traces. The framework supports reproducible comparative analysis and can be extended with richer role taxonomies, larger populations, and repeated-run confidence intervals.

6. Appendix: Reproducibility Notes

- Scenario summary: `results/scenario_summary.csv`
- Comparative figure: `results/graph_papier_recherche.png`
- Per-scenario folders: `results/scenarios/<slug>/`
- Raw conversation logs are preserved in each scenario directory.